FOODHUB ORDER ANALYSIS

Exploratory Data Analysis Report

Business Report

Introduction

FoodHub is a food aggregator company that connects customers with restaurant partners and manages the ordering and delivery process. This report analyzes 1898 orders placed through the FoodHub platform to understand customer preferences, restaurant performance, and the factors that influence order cost, preparation time, delivery time, and customer ratings. The goal is to surface patterns that can help the company improve service and increase revenue.

Data Overview

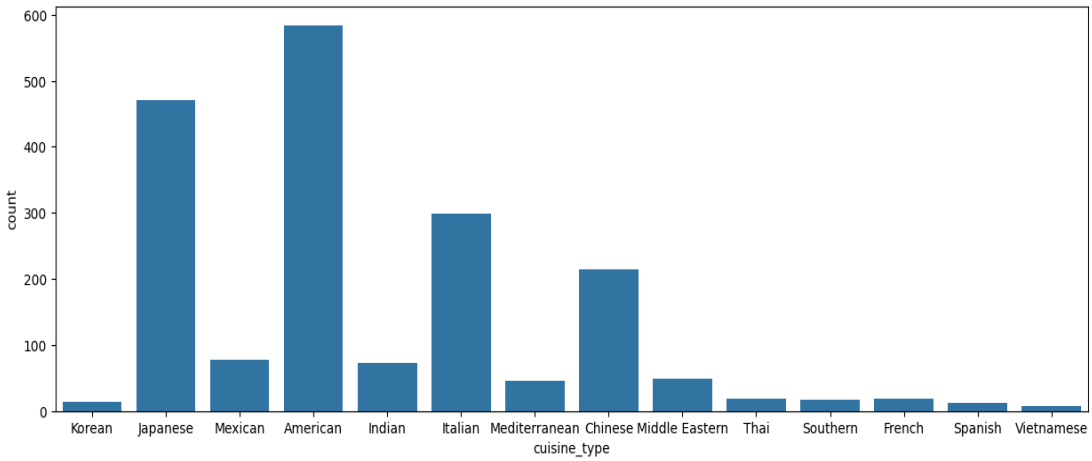
The dataset contains 1898 orders and 9 columns, covering order id, customer id, restaurant name, cuisine type, cost of the order, day of the week, rating, food preparation time, and delivery time. The dataset has 1200 unique customers, 178 unique restaurants, and 14 unique cuisine types. Order id and customer id are numeric identifiers, restaurant name and cuisine type are categorical, and cost of the order, food preparation time, and delivery time are numeric.

Missing Values in Rating

The rating column originally stored ratings as text, mixing the numeric values 3, 4, and 5 with the string "Not given" for orders that were never rated. After converting this column to a proper numeric type, 736 out of 1898 orders (about 39 percent) turned out to have no rating at all. Rather than dropping these rows, they were kept in the dataset so that other patterns, such as cost or cuisine trends, are not affected by discarding a large portion of the data. Only the rating specific analysis excludes these missing values.

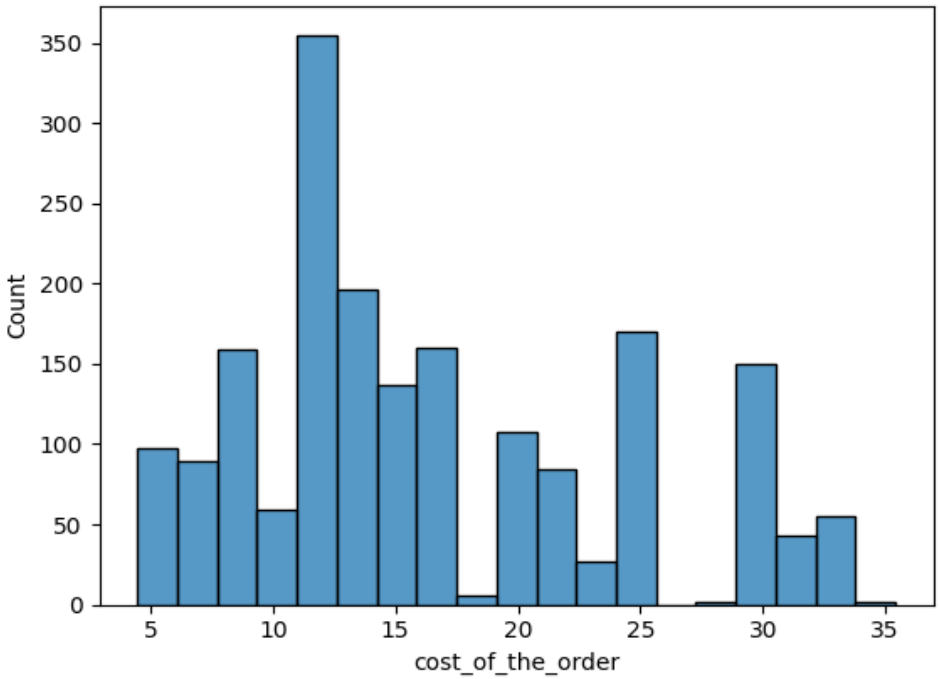
Univariate Analysis

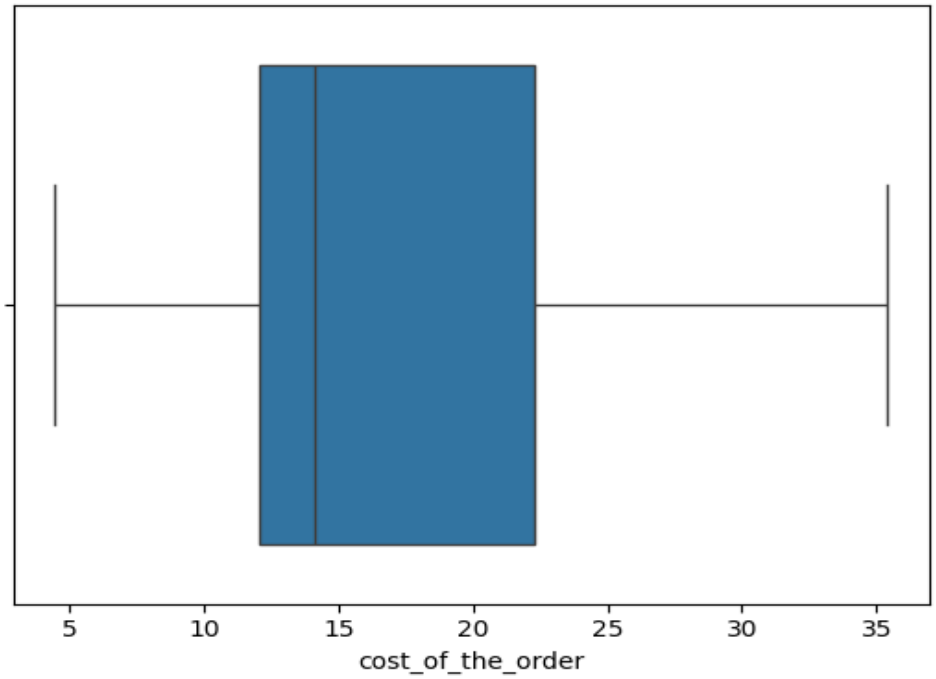
Cuisine Type Distribution



American cuisine is ordered the most, with close to 600 orders, followed by Japanese at around 470 orders. Italian and Chinese cuisines see moderate demand at around 300 and 215 orders respectively. Cuisines such as Vietnamese, Spanish, French, Southern, Thai, and Korean each account for fewer than 25 orders. This shows that customer demand is heavily concentrated in a small number of mainstream cuisines, while regional or niche cuisines make up a very small share of total orders.

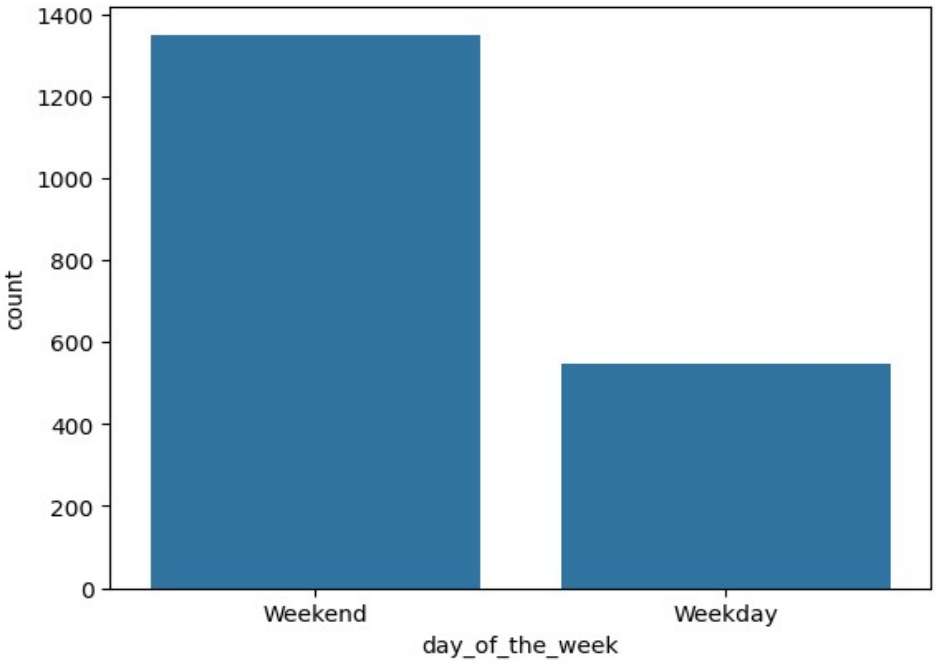
Cost of the Order





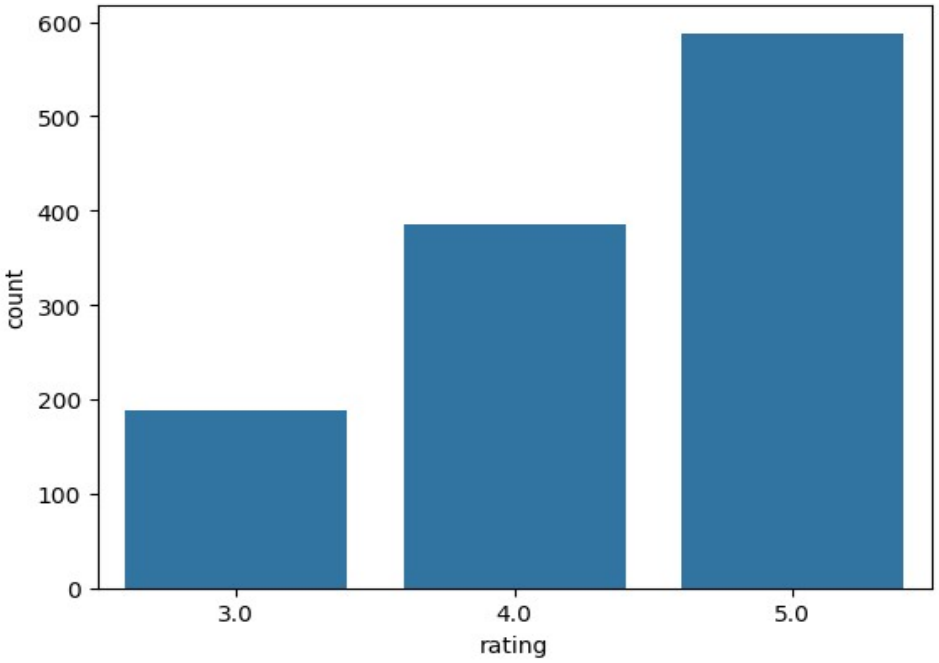
Most orders fall between 10 and 25 dollars, with a sharp concentration around 12 dollars. Very few orders are priced below 5 dollars or above 30 dollars. The boxplot confirms this, showing a median cost of around 14 dollars, with the middle half of all orders falling between 12 and 22 dollars and no outliers present. This indicates that customers on this platform mostly order mid range priced meals, and extreme pricing on either end is rare.

Day of the Week



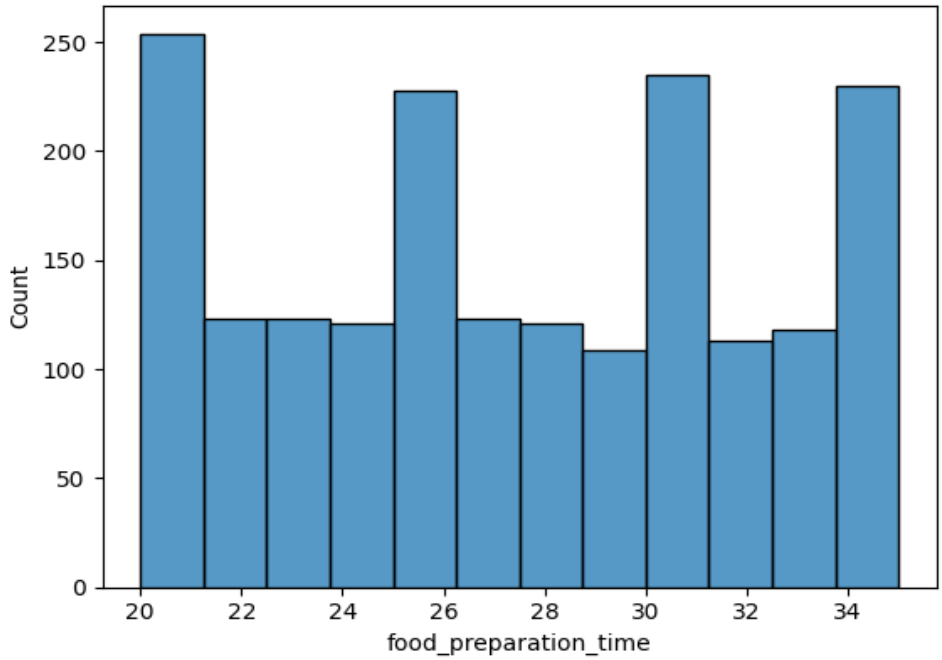
Orders are far more frequent on weekends, at around 1350 orders, compared to weekdays, at around 550 orders. This suggests customers use the platform more when they have free time or are socializing, which is typical of weekend behavior, and points to weekends being the platform's peak demand window.

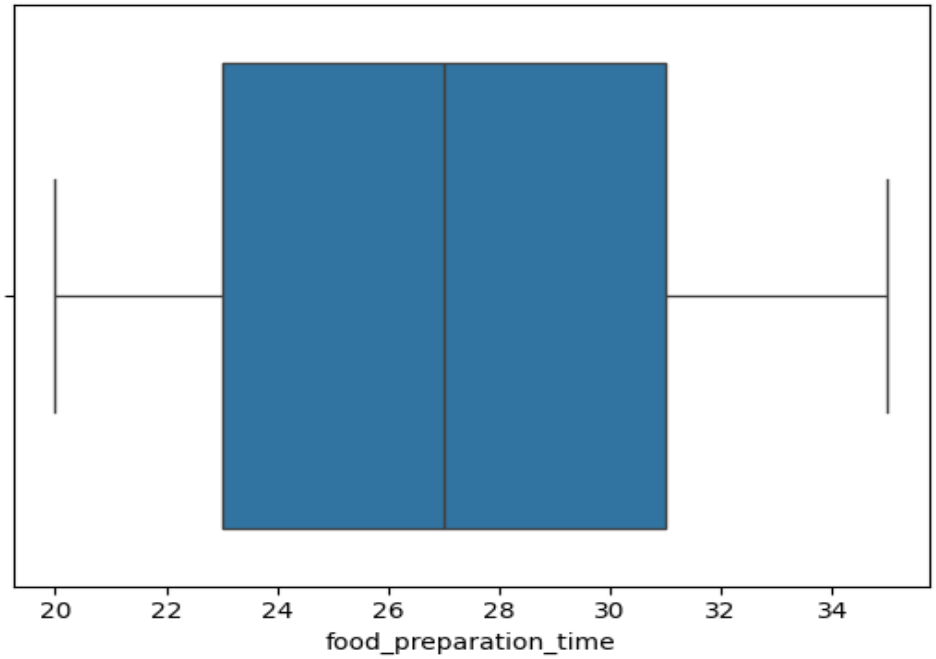
Rating Distribution



Among the orders that were rated, a rating of 5 is the most common, given to around 590 orders, followed by a rating of 4 with around 385 orders. Only around 190 orders received a rating of 3, and no order received a rating below 3. This shows that customer satisfaction on the platform is generally high, since the vast majority of rated orders fall in the top two rating categories.

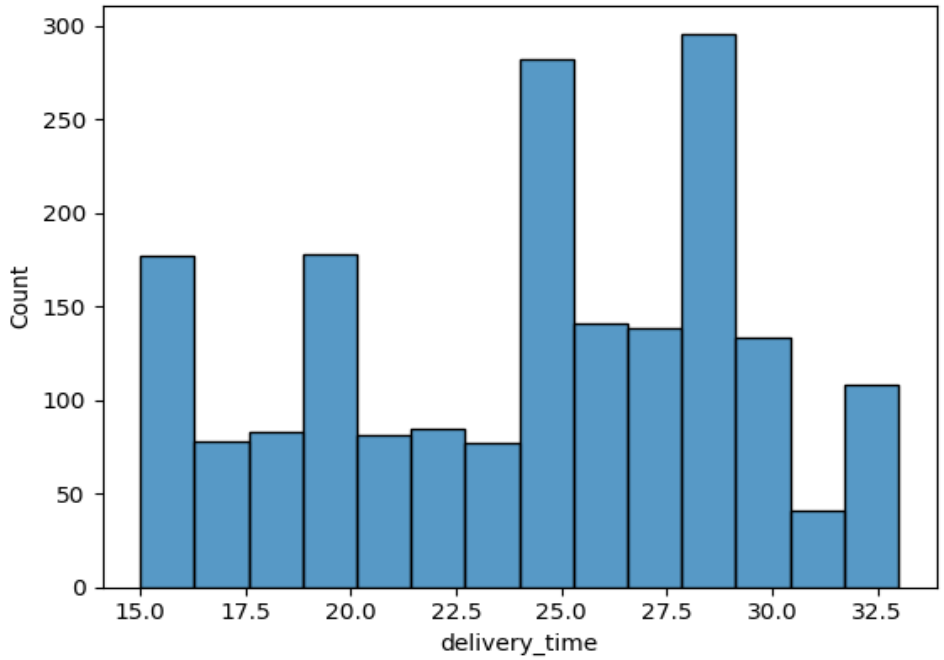
Food Preparation Time

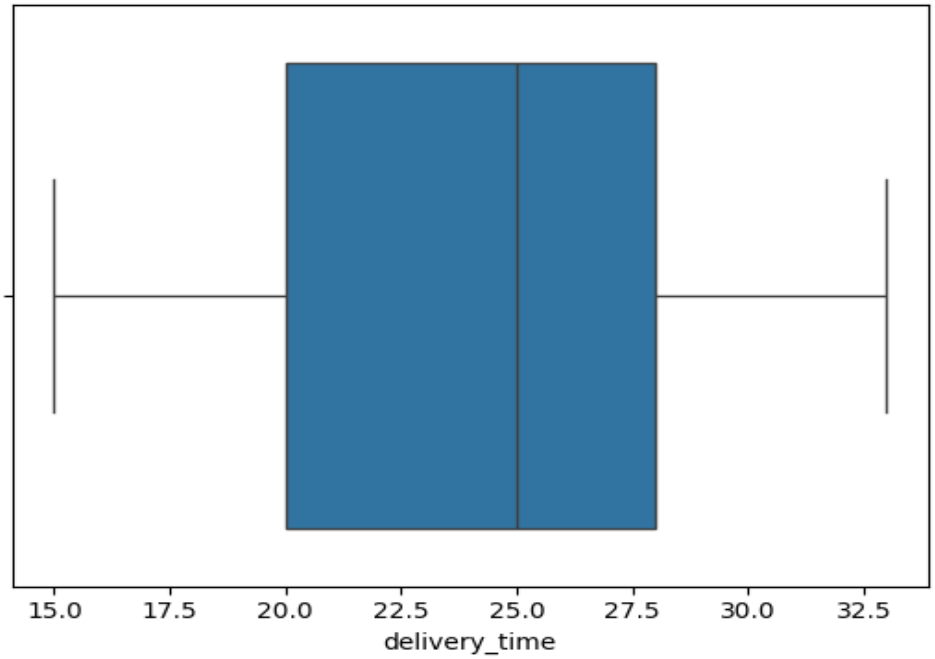




Preparation time is spread fairly evenly between 20 and 35 minutes, without one dominant peak. The boxplot shows a median preparation time of around 27 minutes, with the middle half of orders falling between 23 and 31 minutes and no outliers. This indicates that food preparation is a stable and predictable process across the restaurants on the platform, regardless of cuisine or order cost.

Delivery Time





Delivery times are concentrated between 24 and 29 minutes, with visible peaks around 25 and 28 minutes. The boxplot shows a median delivery time of around 25 minutes, with the middle half of deliveries falling between 20 and 28 minutes and no outliers, suggesting the delivery process is fairly consistent across orders.

Restaurant and Cuisine Insights

Top 5 Restaurants by Number of Orders

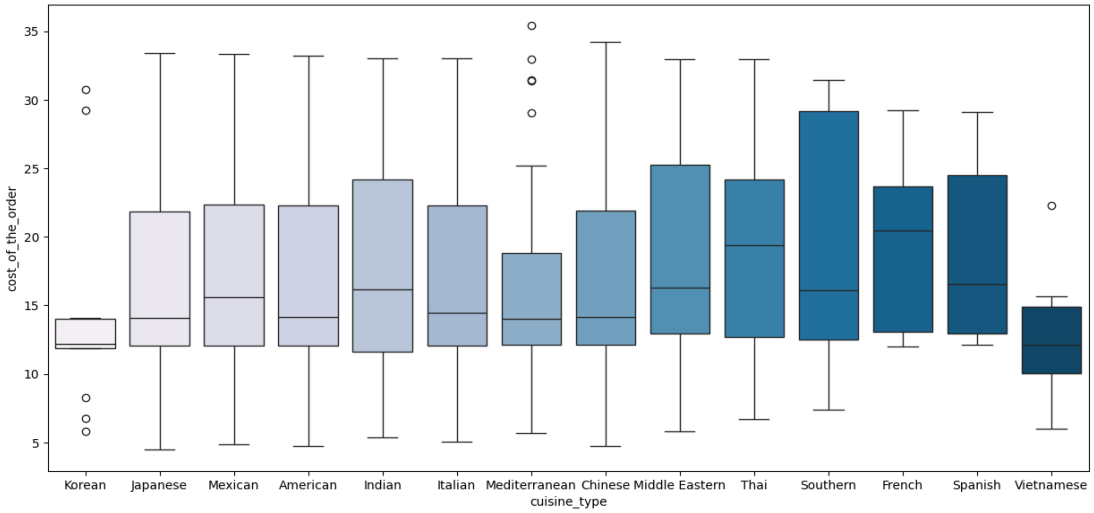
Restaurant	Number of Orders
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Shake Shack is by far the most ordered restaurant on the platform, receiving almost twice as many orders as the second placed restaurant, The Meatball Shop. Blue Ribbon Sushi, Blue Ribbon Fried Chicken, and Parm round out the top five. This shows that a relatively small group of popular restaurants accounts for a large share of total order volume.

Most Popular Cuisine on Weekends

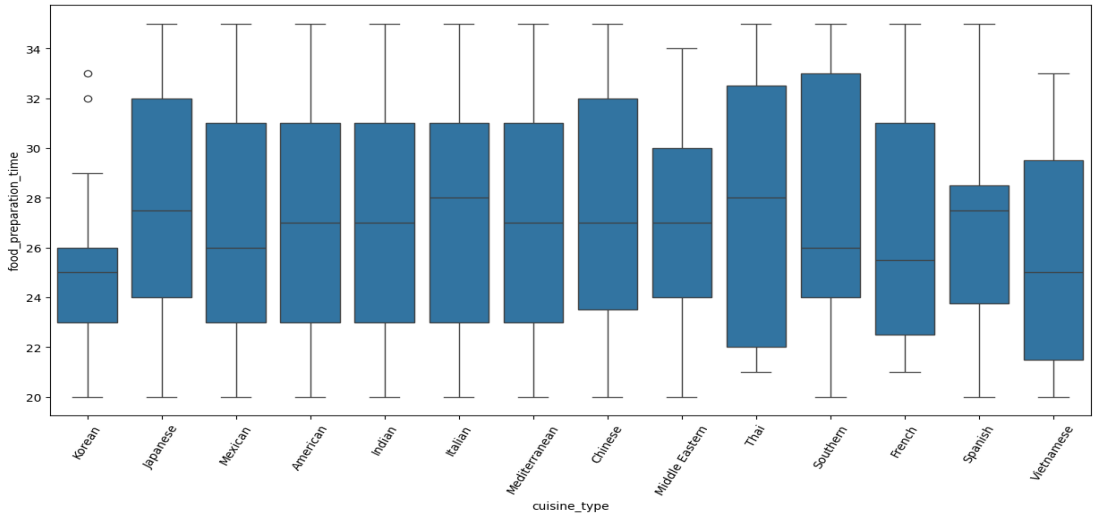
American cuisine leads weekend orders with 415 orders, followed by Japanese with 335 and Italian with 207. Cuisines such as Vietnamese, Korean, Southern, and Spanish each see fewer than 15 weekend orders. This confirms that the same mainstream cuisines that dominate overall demand, American and Japanese in particular, also remain the top choices specifically on weekends.

Cuisine Type versus Cost of the Order



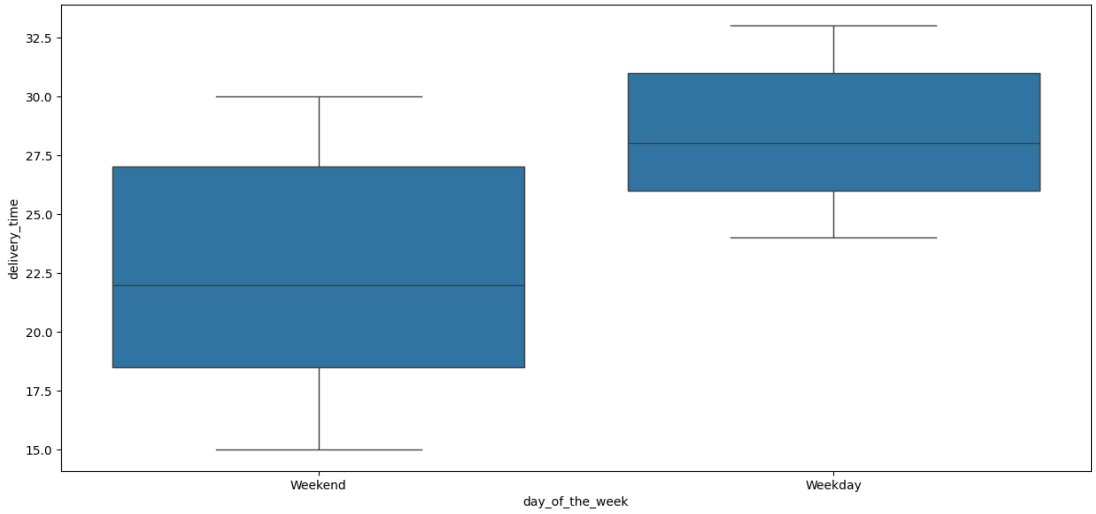
Southern and Middle Eastern cuisines tend to have somewhat higher median order costs, around 16 to 17 dollars, while Korean and Vietnamese cuisines have the lowest medians, around 12 dollars, and a tighter cost range. Korean and Mediterranean cuisines each show a few outliers on the higher end, pointing to occasional high value orders within those categories. Overall, cost varies moderately by cuisine but not dramatically.

Cuisine Type versus Food Preparation Time



Korean cuisine has the lowest median preparation time, at around 25 minutes, though it shows two outliers on the higher side. Most other cuisines, including Thai and Southern, have higher medians of around 26 to 28 minutes and wider ranges. This shows that preparation time varies noticeably across cuisines, likely reflecting differences in how complex each cuisine is to prepare.

Day of the Week versus Delivery Time



Weekday orders have a noticeably higher median delivery time of around 28 minutes, compared to a median of around 22 minutes on weekends. This suggests deliveries are slower on weekdays, most likely due to traffic conditions or delivery staffing being lower relative to weekday demand.

Revenue Generated by Restaurants

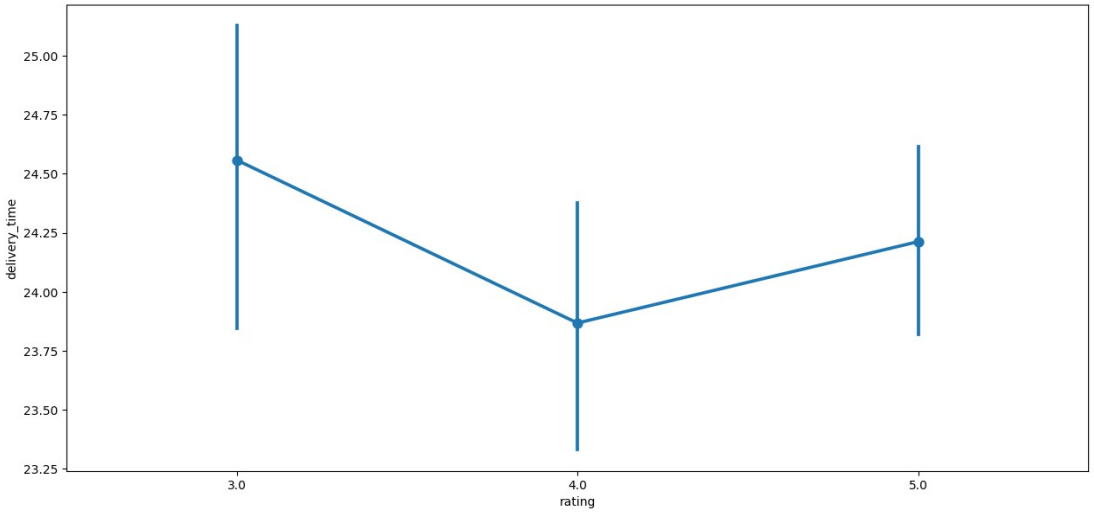
Restaurant	Revenue
Shake Shack	3579.53
The Meatball Shop	2145.21
Blue Ribbon Sushi	1903.95
Blue Ribbon Fried Chicken	1662.29

Parm	1112.76
RedFarm Broadway	965.13
RedFarm Hudson	921.21
TAO	834.50
Han Dynasty	755.29
Blue Ribbon Sushi Bar & Grill	666.62

Shake Shack generates by far the highest revenue among all restaurants, followed by The Meatball Shop and Blue Ribbon Sushi. The remaining restaurants show a gradual decline in revenue. This mirrors the order count ranking, confirming that Shake Shack is the clear revenue leader on the platform, and that a small number of high volume restaurants contribute disproportionately to total revenue.

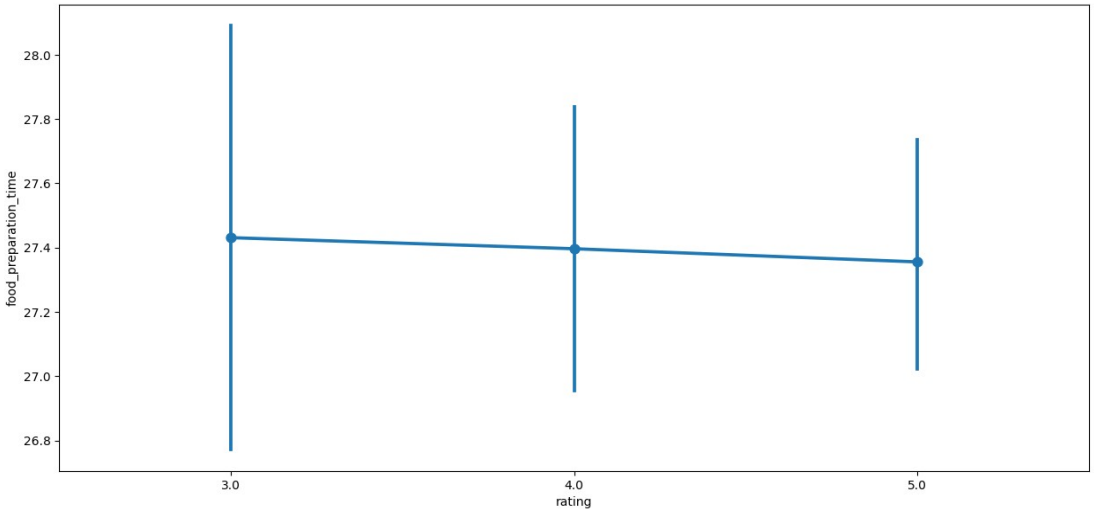
Relationship Between Rating and Other Factors

Rating versus Delivery Time



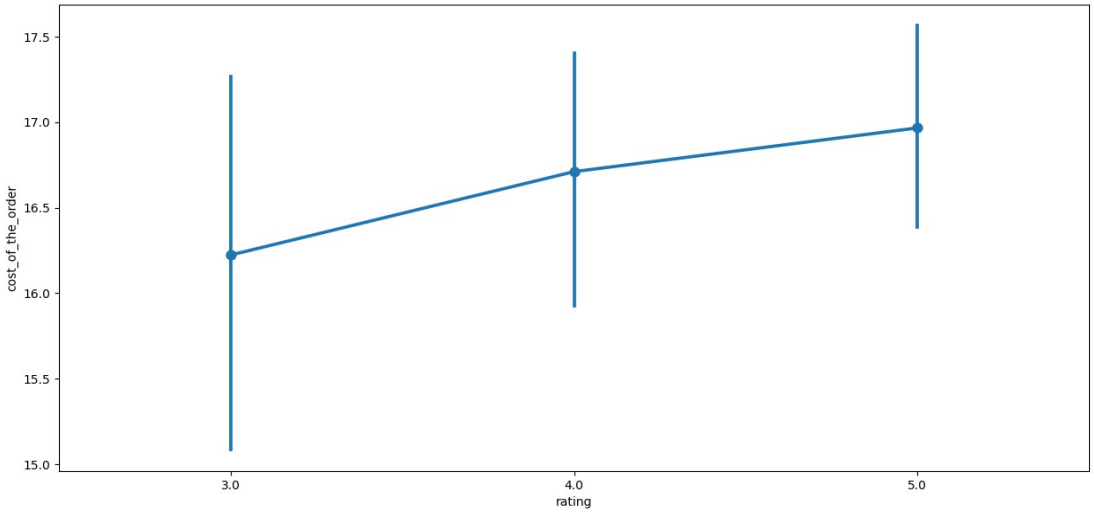
Orders rated 3 have the highest average delivery time, at around 24.5 minutes, while orders rated 4 have the lowest average, at around 23.9 minutes, and orders rated 5 sit around 24.2 minutes. The differences are small and do not follow a clear direction, suggesting that delivery time alone does not strongly determine the rating a customer gives.

Rating versus Food Preparation Time



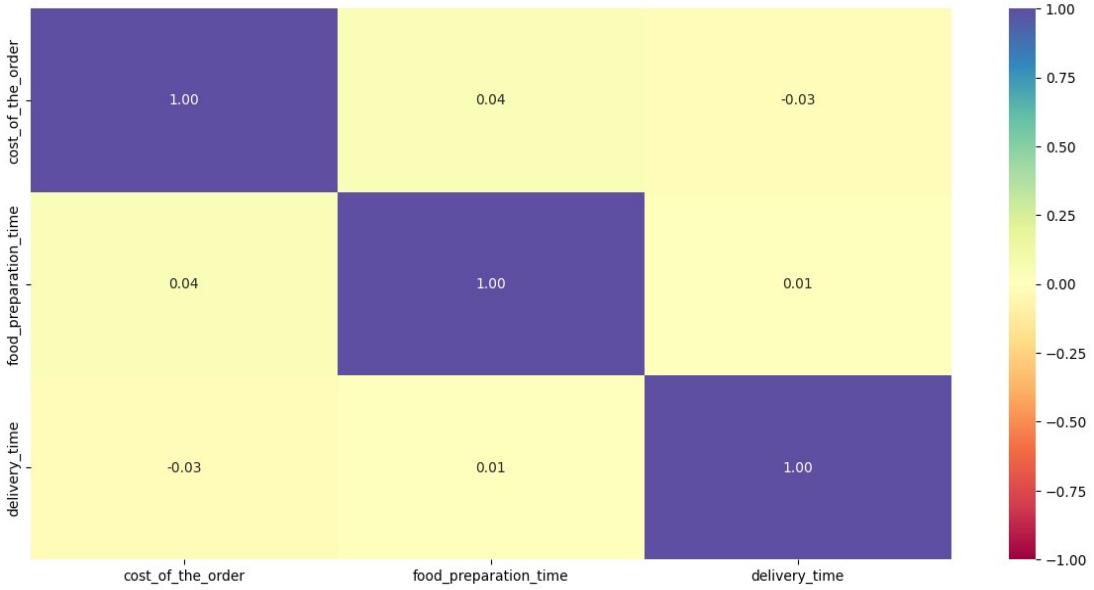
Average food preparation time stays almost the same across all rating levels, ranging narrowly from about 27.3 to 27.4 minutes. This indicates that food preparation time has little to no impact on the rating customers give to their order.

Rating versus Cost of the Order



Average order cost rises slightly as rating increases, from about 16.2 dollars for a rating of 3 to about 17.0 dollars for a rating of 5. This suggests customers placing higher priced orders tend to give marginally better ratings, though the effect is small.

Correlation Among Numeric Variables



The correlation values between cost of the order, food preparation time, and delivery time are all close to zero, ranging from negative 0.03 to 0.04. This means none of these three variables have a meaningful linear relationship with one another. A costlier order does not take longer to prepare or deliver, preparation time does not influence delivery time, and delivery time is not linked to how expensive an order is.

Since these three operational factors work independently, delays or inefficiencies in one process, such as food preparation, are not caused by or connected to the other processes. This means pricing, kitchen operations, and delivery logistics should each be evaluated and optimized on their own rather than assuming an improvement in one area will carry over to another.

Additional Business Metrics

Overall Delivery Performance

The average delivery time across all orders is about 24.16 minutes. When food preparation time and delivery time are combined, about 10.54 percent of all orders take more than 60 minutes from order placement to delivery, which represents a meaningful share of orders where the total wait time may affect the customer experience.

High Value Orders

555 orders, or about 29.24 percent of all orders, are priced above 20 dollars. This is a sizeable segment of higher value transactions that contributes disproportionately to revenue given the revenue formula applied by the company.

Top Customers

Customer 52832 placed the highest number of orders at 13, followed by customer 47440 with 10 orders and customer 83287 with 9 orders. These repeat customers represent a small but loyal group that could be targeted for retention efforts such as loyalty rewards.

Restaurants with Strong and Consistent Ratings

Among restaurants that received more than 50 ratings, four stand out with an average rating above 4: The Meatball Shop at 4.51, Blue Ribbon Fried Chicken at 4.33, Shake Shack at 4.28, and Blue Ribbon Sushi at 4.22. These restaurants combine high order volume with consistently strong customer satisfaction, making them strong candidates for promotional partnerships.

Total Revenue

Applying the company's revenue formula, where orders above 20 dollars earn 25 percent commission, orders between 5 and 20 dollars earn 15 percent commission, and orders of 5 dollars or below earn no commission, the total revenue generated across all orders in the dataset is around 6166.30 dollars.

Conclusions and Recommendations

- American and Japanese cuisines drive the bulk of demand, so the company should prioritize expanding restaurant partnerships and marketing within these categories while still maintaining a presence in niche cuisines for variety.
- Weekends are the peak ordering period, so restaurant staffing, delivery fleet availability, and app promotions should be scaled up specifically around weekends to match demand.
- Weekday deliveries are slower than weekend deliveries, which is worth investigating further, since this is unusual and may point to staffing gaps or traffic patterns on weekdays that could be addressed with better delivery planning.
- A small number of restaurants, led by Shake Shack, account for a disproportionate share of both orders and revenue, so retaining strong relationships with these top partners should be a priority.
- About 10.54 percent of orders take more than 60 minutes in total, which is worth monitoring, since consistently long wait times can hurt customer satisfaction even though overall ratings are currently high.
- Cost, food preparation time, and delivery time operate independently of each other, so improvement initiatives targeting one of these areas, such as faster delivery, should be planned and measured separately rather than assuming gains will spill over into the other areas.
- Restaurants such as The Meatball Shop, Blue Ribbon Fried Chicken, Shake Shack, and Blue Ribbon Sushi combine high order volume with strong average ratings, making them good candidates to feature in app recommendations or promotional campaigns.